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SUPERSEDING
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MILITARY SPECIFICATION

WATTMETER, TS-118( )/AP

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a portable test set, designated as Wattmeter TS-118( )/AP, for measuring average radio frequency (rf) power within the frequency range of 20 Mc to 1400 Mc. (See 6.1 and 6.4).

1.2 Wattmeter TS-118( )/AP consists of the following units:

Dummy Load, Electrical DA-140( )/AP.
Thermocouple, Heating Element MX-1781( )/U.
Thermocouple, Heating Element MX-1782( )/U.
Thermocouple, Heating Element MX-1783( )/U.
Thermocouple, Heating Element MX-1785( )/U.
Shunt, Tuning MX-1784( )/U.
Adapter, Connector UG-83B/U.
Adapter, Connector RF, UG-213A/U.
Adapter, Connector RF, UG-999( )/U.
Cable Assembly, RF, CG-92F/U(-ft-in)
Case, Transit CY-1764( )/UPM.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
UU-T-111	Tape; Paper, Gummed (Sealing and Securing)
PPP-B-636	Boxes, Fiber

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PPP-F-00320	Fiberboard, Sheet, Stock and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior, Packaging.
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant, of Communications Electronic, and Associated Electrical Equipment.
MIL-T-945	Test Equipment, for Use with Electronic Equipment; General Specification.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-R-17252	Rubber, Cellular (Bonded Shredded).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108	Definitions of, and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-170	Moisture Resistance Test Cycle.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for.

DRAWINGS

ELECTRONICS COMMAND

SC-D-23249	Cable Assembly, RF, CG-92F/U (-ft -in)
SC-A-46439	List of Accessories for Package Tester
SC-DL-334850	Thermocouple, Heating Element MX-1785( )/U
SC-DL-334851	Thermocouple, Heating Element MX-1781( )/U
SC-DL-334852	Thermocouple, Heating Element MX-1782( )/U
SC-DL-334853	Thermocouple, Heating Element MX-1783( )/U

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SC-DL-334854	Shunt, Tuning MX-1784( )/U
SC-DL-334855	Adapter, Connector, RF, UG-999( )/U.
SC-DL-334857	Dummy Load, Electrical DA-140( )/AP.
SC-DL-334858	Case, Transit CY-1764( )/UPM

NAVY

REB49274	Adapter, Connector UG-21^{2}A/U
REA49302	Adapter, Connector UG-83^{3}B/U
10-T-2145	Light Weight Stock Testing Machine.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification.

(Application for copies of the above publications should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

U. S. POST OFFICE DEPARTMENT

Parcel Post Regulations

(Application for copies should be obtained from any first class U. S. Post Office).

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3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with drawings listed in 3.2 and its subparagraphs.

3.2 Units.

3.2.1 Dummy Load, Electrical, DA-140( )/AP. Dummy load, electrical DA-140( )/AP shall be constructed in accordance with Drawing and Data List SC-DL-334857.

3.2.2 Thermocouples.

3.2.2.1 Thermocouple, Heating Element MX-1781( )/U. Thermocouple, Heating Element MX-1781( )/U shall be constructed in accordance with Drawing and Data List SC-DL-334851.

3.2.2.2 Thermocouple, Heating Element MX-1782( )/U. Thermocouple, Heating Element MX-1782( )/U shall be constructed in accordance with Drawing and Data List SC-DL-334852.

3.2.2.3 Thermocouple, Heating Element MX-1783( )/U. Thermocouple, Heating Element MX-1783( )/U shall be constructed in accordance with Drawing and Data List SC-DL-334853.

3.2.2.4 Thermocouple, Heating Element MX-1785( )/U. Thermocouple, Heating Element MX-1785( )/U shall be constructed in accordance with Drawing and Data List SC-DL-334850

3.2.3 Shunt, Tuning MX-1784( )/U. Shunt, Tuning MX-1784( )/U shall be constructed in accordance with Drawing and Data List SC-DL-334854.

3.2.4 Adapter, Connector UG-83B/U. Adapter, Connector UG-83B/U shall be constructed in accordance with Drawing REA49302.

3.2.5 Adapter, Connector UG-213A/U. Adapter, Connector UG-213A/U shall be constructed in accordance with Drawing REB49274.

3.2.6 Adapter, Connector, RF, UG-999( )/U. Adapter, Connector, RF, UG-999( )/U shall be constructed in accordance with Drawing and Data List SC-DL-334855.

3.2.7 Cable Assembly, RF, CG-92F/U (-ft -in). Cable Assembly, RF, CG-92F/U (-ft -in) shall be constructed in accordance with Drawing SC-D-23249.

3.2.8 Case, Meter CY-1764( )/UPM. Case, Transit CY-1764( )/UPM shall be constructed in accordance with Drawing and Data List SC-DL-334858.

3.3 Processes.

3.3.1 Processes shall conform to Specification MIL-T-945.

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3.3.2 Tropicalization.

3.3.2.1 Treatment of Assembled Equipment. Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.
(2) It does not incorporate any material listed in 3.3.2.2 unless the material has been given the treatment specified in Specification MIL-T-945 or a treatment approved by the contracting officer.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-T-945.

3.3.2.2 Materials not resistant to moisture and fungi. The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair, or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.3.2.3 Statement of treatment. The contractor shall submit for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. This treatment shall include the statements required by Specifications MIL-T-945 and MIL-T-152, as applicable.

3.3.3 Marking. Marking shall conform to Specification MIL-M-13231. (See 4.11).

3.3.4 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.11)

3.4 Electrical requirements.

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3.4.1 Frequency and power range. Radio Frequency Wattmeter TS-118( )/AP shall be capable of measuring indications of power output over the following power and frequency ranges:

Frequency range	Power
20 - 500 mc	2 - 500 Watts
500 - 1000 mc	2 - 300 Watts
1000 - 1400 mc	2 - 225 Watts

The instrument shall have a full scale reading of 500 watts from 20 mc to 500 mc; thereafter the full scale reading shall decrease from 500 watts at 500 mc, to 300 watts at 1000 mc, and then to 225 watts at 1400 mc.

3.4.1.1 Accuracy of indicated power. The dc millivoltmeter shall be so calibrated that when used with a thermocouple marked with its predetermined sensitivity correction factor (C factor) as specified in 3.4.4.1, and calibration curves covering the frequency range of 20 mc to 1400 mc, the power output shall be determined with an accuracy of within ±15 percent of the absolute.

3.4.2 Dummy Load, Electrical DA-140 ( )/AP. Dummy Load, Electrical DA-140( )/AP shall be a 500-watt coaxial dummy load with a radio frequency (rf) input impedance of 50 ohms.

3.4.2.1 Dc resistance. The dc resistance of the Dummy Load, Electrical DA-40( )/AP shall be 50 ohms ±3 percent.

3.4.2.2 The cold resistance measured before and after the 500-watt test (see 4.6.1) shall not vary by more than ±0.2 ohms.

3.4.3 Impedance match. The voltage standing wave ratio (VSWR) of Dummy Load, Electrical DA-140( )/AP used in conjunction with Adapter, Connector UG-999( )/U shall not exceed 1.25 to 1 over the frequency range of 20 to 1400 mc.

3.4.3.1 VSWR, (combination). The voltage standing wave ratio, (VSWR, of Dummy Load, Electrical, DA-140( )/AP; Adapter, Connector UG-999( )/U; Shunt, Tuning MX-1784( )/U; and each of the individual thermocouples shall not exceed the applicable value listed in Table A over the frequency range of 20 - 1400 mc.

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Table A. VSWR limits

Combination	VSWR
Load DA-140( )/AP + Adapter UG-999( )/U + Shunt MX-1784( )/U + Thermocouple MX-1781( )/U.	1.55
Load DA-140( )/AP + Adapter UG-999( )/U + Shunt MX-1784( )/U + Thermocouple MX-1782( )/U	1.30
Load DA-140( )/AP + Adapter UG-999( )/U + Shunt MX-1784( )/U + Thermocouple MX-1783( )/U	1.35
Load DA-140( )/AP + Adapter UG-999( )/U + Shunt MX-1784( )/U + Thermocouple MX-1785( )/U	1.75

3.4.4 Thermocouples.

3.4.4.1 Sensitivity correction factor. The thermocouples shall have a "C" factor (sensitivity figure) calibration stamped on the body of the thermocouple. The "C" factor may vary from 0.900 to 1.1000, but the calibration shall be made to the nearest 0.01.

3.4.4.2 Operating range. Full scale deflection on the meter for specific thermocouples shall be obtained at 400 mc with the following power outputs:

Thermocouple, heating element	Power	Power output tolerance
MX-1785( )/U.....	18 watts	± 10%
MX-1781( )/U.....	46 watts	± 10%
MX-1782( )/U.....	180 watts	± 10%
MX-1783( )/U.....	560 watts	± 10%

3.4.4.3 VSWR, (Thermocouple). The voltage standing wave ratio, (VSWR), of individual Thermocouples MX-1781( )/U, MX-1782( )/U, MX-1783( )/U and MX-1785( )/U in combination with Dummy Load DA-140( )/AP, Adapter, Connector UG-999( )/U and Tuning Shunt MX-1784( )/U shall not exceed the values listed in Table A at any frequency between 20 mc and 1400 mc, (See 4.6.4.1).

3.4.4.4 Thermocouple accuracy of indicated power. The accuracy of the indicated power of the individual thermocouple in combination with Load DA-140( )/AP, Adapter UG-999( )/U, and Shunt MX-1785( )/U shall be within ±15 percent of the actual power output at any frequency between 20 mc and 1400 mc. (See 4.6.6).

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3.4.5 Shunt, Tuning MX-1784( )/U.

3.4.5.1 VSWR (Tuning shunt). The VSWR of the Tuning Shunt MX-1784( )/U in combination with Dummy Load, DA-140( )/AP, Adapter, Connector UG-999( )/U and Thermocouple MX-1781( )/U, MX-1782( )/U, MX-1783( )/U, or MX-1785( )/U shall not exceed the values listed in table A at any frequency between 20 mc and 1400 mc. (See 4.6.4.2)

3.4.5.2 Shunt accuracy of indicated power. The accuracy of the indicated power of the tuning shunt in combination with Load DA-140( )/AP, Adapter UG-999( )/U and Thermocouple MX-1781/U, MX-1782/U, MX-1783/U, or MX-1785/U shall be within ± 15 percent of the actual power output at any frequency between 20 mc and 1400 mc. (See 4.6.7).

3.5 Service conditions. The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.5.1 Temperature (see 4.7.4).

(a) Operating: Ambient temperature in the range of +125°F to -4°F. Exposure of the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.5.2 Relative humidity (see 4.7.5). Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.5.3 Elevation (see 4.7.4).

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.5.4 Immersion (see 4.7.3). Three feet of water for 2 hours, (in transit case).

3.5.5 Orientation.

(a) Operating (see 4.6.5.4). The equipment shall operate in any position except those in which the load resistor axis is within 60° of the vertical, connector-up position, or 45° of the vertical, connector-down position.

(b) Nonoperating: Storage in any position for an indefinite period. Approval of the preproduction sample shall be considered as compliance with this requirement.

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3.6 Vibration, bounce, and shock.

3.6.1 Vibration. Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked or removed. (See 4.7.6).

3.6.2 Bounce and shock. The equipment shall meet the requirements of table I.

Table I. Bounce and shock

Requirement	Inspection paragraph	Performance after inspection
Bounce test	4.7.1	Full specification performance. No physical damage except surface abrasion.
Shock. Drop	4.7.2	Equipment shall be operable. Minor mechanical damage is permissible (note A).
(Hammer)	4.7.2.1	

Note A. After this shock test, equipment performance shall conform to the following: Accuracy of indicated power (see 4.6.5).

3.7 Preconditioning. IStems shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.8 Preproduction samples. The contractor shall furnish two (2) preproduction samples for approval. (See 4.2.)

3.9 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order (see 6.2).

3.10 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing.

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3.11 Workmanship. The item shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL-T-945:
General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connectors, general.
Grounding, general.
Shielding on wire and cable.
Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process of welding.
Mechanical operation of controls.
Gears.
Self-tapping screws.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial Laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.2).

(b) Acceptance inspection. (Acceptance inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3, 4.8, and 4.9.

(1) Acceptance inspection of items before preparation for delivery (see 4.3 and 4.8).

(2) Acceptance inspection of preparation for delivery (see 4.9 and section 5).

4.2 Preproduction inspection. This inspection shall consist of the preproduction inspection specified in table II, the inspection specified in the subsidiary documents covering items listed in 4.8, and the group A and group B inspection specified in tables III and IV, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, drop, and (4) watertightness; other preproduction inspection may precede, follow, or be interspersed between the foregoing. The sampling plan and indicated AQL's do not apply for preproduction inspection.

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Table II. Preproduction inspection

Inspection test	Requirement paragraph	Inspection paragraph
Elevation	3.5.3	4.7.5
Orientation	3.5.5.	4.6.5.4
Temperature	3.5.1	4.7.5
Humidity	3.5.2	4.7.6
Vibration	3.6.1	4.7.7
Bounce	3.6.2	4.7.1
Shock, drop (hammer)	3.6.2	4.7.2
Immersion(watertightness)	3.5.4	4.7.3

4.3 Acceptance inspection of items before preparation for delivery. The contractor shall perform the inspection specified in 4.8 and 4.3.1 through 4.3.3. Each unit which will be subjected to group A or group B inspection shall be preconditioned after final assembly. (See 3.7) Test equipment, when required for Government verification inspection, shall be made available by the contractor.

4.3.1 Group A inspection. This inspection (including sampling) shall conform to table III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table III. Group A inspection

Inspection	Requirement paragraph	Inspection paragraph	A Q L	
			Major	Minor
Visual and mechanical examinations	3.2.2.1 through 3.2.2.4	4.5	1.0%	4.0%
Wattmeter TS-118( )/AP Thermocouple units				
Adapters, Connector	3.2.4 through 3.2.6	4.5	1.0%	4.0%
Shunt, tuning	3.2.3	4.5	1.0%	4.0%
Dummy load DA-140( )/AP	3.2.1	4.5	1.0%	4.0%
Cable Assembly CC-92(F)/U(-ft-in)	3.2.7	4.5	1.0%	4.0%
Case, Transit	3.2.8	4.5	1.0%	4.0%
Electrical tests				
Dummy Load 500-watt test	3.4.2	4.6.1	1.5 dphu for combined group	*
Impedance match of load	3.4.3	4.6.2		
VSWR	3.4.3.1	4.6.4		
Thermocouple sensitivity "C" factor	3.4.4.1	4.6.3	1.0%	*
Accuracy of indicated power	3.4.1.1	4.6.5 and 4.6.5.4	1.0%	*
Thermocouple VSWR	3.4.4.3	4.6.4.1	1.0%	*
Thermocouple Accuracy of Indicated Power	3.4.4.4	4.6.6	1.0%	*
Shunt VSWR	3.4.5.1	4.6.4.2	1.0%	*
Shunt Accuracy of Indicated Power	3.4.5.2	4.6.7	1.0%	*

*All electrical defects are considered major.

4.3.2 Group B inspection. This inspection, including sampling, shall conform to table IV and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.3.2.1 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

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Table IV. Group B inspection

Inspection	Requirement paragraph	Inspection paragraph
Watertightness	3.5.4	4.7.3
Interchangeability	3.10	4.8

4.3.3 Reinspection of conforming group B sample units. Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.4 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the package tester as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.5 Visual and mechanical inspection. Equipment shall be examined for defects listed in Standard MIL-STD-252, and Table V.

Table V. Visual and mechanical defects

Classification	Defect
Major	Carry handle, hinge, or lid fastener improperly attached. Cover missing or inoperative.
Minor	Carry handle or cover difficult to operate

4.6 Electrical tests.

4.6.1 Dummy load electrical 500-watt test. Several Dummy Loads, Electrical DA-140( )/AP shall be connected as shown in figure 2. Each unit, less indicating meter shall be subjected to a heating run at 60 cycles 500 watts for a period of 20 minutes. The dc resistance when measured with any instrument having an accuracy of measurement to within 1/10 of 1 percent, shall conform to 3.4.2.

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4.6.2 Impedance match of load. Using the slotted line test circuit shown in figure 3, the VSWR of each Dummy Load, Electrical DA-140( )/AP, in conjunction with Adapter, Connector JG-999( )/U shall be measured to determine compliance with 3.4.3. Measurements shall be made at 50 mc, 1000 mc, 1200 mc, 1300 mc, 1350 mc, and 1400 mc, respectively. The VSWR shall be read directly from the indicator shown in the test circuit.

4.6.3 Thermocouple sensitivity correction (C) factor. Using any of the test circuits shown in figures 4A, 4B, or 4C, the thermocouples shall be tested to determine compliance with 3.4.4.1.

4.6.3.1 Standard thermocouple test circuit. If the test circuit shown on figure 4A is utilized, the following power requirements shall be applied to each type of thermocouple:

Thermocouple, heating element	Power requirement
MX-1785( )/U	9 watts
MX-1781( )/U	25 watts
MX-1782( )/U	90 watts
MX-1783( )/U	250 watts

The "C" factor shall be determined by the following formula:

$$\text{"C" factor} = \frac{\text{Standard Reading}}{\text{Thermocouple under test}}$$

4.6.3.2 Slotted line test circuit. If the test circuit using a pass through slotted line, or a wattmeter calibrated for power measurements is used as shown in figure 4B, the "C" factor shall be determined in the following manner:

$$\text{"C" factor} = \frac{\text{Power Standard}}{\text{Indicated Power}}$$

4.6.3.3 Low frequency test circuit. If the low frequency test circuit shown on figure 4C is employed, comparing the output of the thermocouple under test with that of a standard thermocouple, the "C" factor shall be determined in accordance with the following formula:

$$\text{"C" factor} = \frac{\text{Standard Indication}}{\text{Thermocouple under test}}$$

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4.6.4 VSWR test. The equipment shall be connected as shown on figure 5, and shall be tested to determine compliance with 3.4.3.1. The VSWR shall be determined at 20 mc, 1,000 mc, 1,200 mc, 1,300 mc, 1,350 mc, and 1,400 mc. Random combinations of thermocouples and loads shall be used to make the measurements.

4.6.4.1 Thermocouple VSWR test. The thermocouple under test shall be connected as shown on figure 4B, and shall be tested to determine compliance with 3.4.4.3. The VSWR shall be determined at 20, 1000, 1200, 1300, and 1400 mc.

4.6.4.2 Shunt VSWR test. The Tuning Shunt MX-1784( )/U under test shall be connected as shown on figure 4B, and shall be tested to determine compliance with 3.4.5.1. The VSWR shall be determined at 20, 1000, 1200, 1300, and 1400 mc.

4.6.5 Accuracy of indicated power. The test circuit shall be connected as shown on figure 6 and the equipment shall be tested to determine compliance with 3.4.1.1. The pass through wattmeter used in the test circuit shall be calibrated against a primary standard.

4.6.5.1 Tests shall be made with thermocouple MX-1785( )/U and MX-1781( )/U at those power outputs which indicate full scale deflection at the nominal frequencies of 50 mc, 400 mc, 1,000 mc, and 1,400 mc, respectively.

4.6.5.2 Thermocouple MX-1783( )/U shall be tested at nominal frequencies of 200 mc and 400 mc at the 500-watt setting. It shall then be tested at the nominal frequencies of 1,000 mc and 1,400 mc at a power output which gives full scale deflection.

4.6.5.3 Thermocouple MX-1782( )/U shall be checked at a power indication of 400 watts at the nominal frequency of 400 mc. It shall then be tested at the nominal frequencies of 1,000 mc and 1,400 mc at those power outputs which give full scale defection.

4.6.5.4 The accuracy of indicated power shall also be determined when the test set is oriented as specified in 3.5.5.

4.6.6 Thermocouple test, accuracy of indicated power. The thermocouple under test shall be connected as shown on figure 6. Using the calibration curves and at a power output which will give full scale meter deflection, the thermocouple shall be tested for compliance with 3.4.4.4.

4.6.7 Tuning Shunt, Accuracy of indicated power. The shunt under test shall be connected as shown on figure 6. Using the calibration curves and at a power output which will give full scale meter deflection, the shunt shall be tested for compliance with 3.4.5.2.

4.7 Service condition tests.

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4.7.1 Bounce test, (See 3.6.2). The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment in its transit case prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the back-stop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.7.2 Shock test. In order to determine compliance with 3.6.2, the TS-118( )/AP shall be tested in accordance with procedure specified in 4.7.2.1.

4.7.2.1 Shock test machines. The shock testing machine shall be as shown on Navy Drawing 10T-2145. The test set shall be mounted in such a way as to approximate the actual rigidity encountered in service in the utilization of the particular test set. Portable test sets without transit cases shall be secured to the test machine by means of canvas straps. The test set shall also be tested de-energized in the transit case.

4.7.2.2 Test. A total of nine blows shall be applied. Three blows shall be applied parallel to each of the three principal axes of the test set being tested. The height of hammer drop for the back blow and side blow shall be 1 foot, 2 feet, and 3 feet. For the top blow, the height of hammer drop shall be 2 feet, 3 feet, and 4 feet. After each blow of the shock test, the test set shall continue to perform its principal functions without any attention to or replacement of parts. Adjustment will be permitted to external controls only. Momentary failure to operate during shock is permissible providing the test resumes normal operation after each blow. Minor damage or distortion will be permitted, provided such damage or distortion cannot in any way impair the ability of the test set to perform its principal functions.

4.7.3 Watertightness. Transit Case CY-1764( )/UPM shall comply with the requirements of 3.5.4 when tested in accordance with Standard MIL-STD-108.

4.7.4 Temperature and altitude tests. The following tests shall be conducted with the test set in a temperature-altitude chamber in the order specified. Suitable provisions shall be made to pass the required electrical connections through the walls of the chamber. To check temperature stabilization, place a thermocouple in good thermal contact on the largest internal mass centrally located or in any other location deemed necessary by the Government inspector.

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(a) With the test set stabilized at an ambient temperature of 25°C (77°F), the test set shall be operated for a 20-minute warmup period, to determine compliance with full specified performance.

(b) With the test set still operating, the ambient temperature shall be lowered to -20°C (-4°F) and sufficient time for temperature stabilization shall be allowed.

(c) It shall be determined whether the performance is maintained and all controls and switches can be readily manipulated.

(d) With the test set not operating the ambient temperature shall be lowered to -62°C (-80°F) and sufficient time for temperature stabilization shall be allowed.

(e) The ambient temperature shall be raised to 25°C(77°F) and temperature shall be held at ambient until stabilization has been reached.

(f) The barometric pressure shall be lowered to 20.6 inches of mercury. After a 15-minute period it shall be determined whether the full performance requirements of this specification are maintained.

(g) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(h) The pressure shall be raised to ambient atmospheric and the test set shall be operated to determine whether the full performance requirements of this specification are maintained.

(i) With the test set still operating, the ambient temperature shall be raised to 52°C (125°F) and sufficient time for temperature stabilization shall be allowed.

(j) The test set shall be operated for 30 minutes, and it shall be determined whether the full performance requirements of this specification are maintained.

(k) With the test set not operating, the ambient temperature shall be raised to 71°C (160°F) and sufficient time for temperature stabilization shall be allowed.

(l) The ambient temperature shall be lowered to 25°C (77°F) and held for 30 minutes after temperature stabilization. The test set shall be inspected for degradation.

(m) The test set shall be turned on for sufficient time to determine whether the set will function satisfactorily. Then the test set shall be turned off to allow temperature stabilization. This procedure shall be repeated three times.

(n) The test set shall be operated to determine whether the full performance requirements in this specification are maintained.

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4.7.5 Humidity test. The test set shall be subjected to the following:

a. Dry the test set for 24 hours at 130° ±5°F.

b. Measure performance of the test set for compliance with 3.4.1.1 and make any necessary adjustments to enable the test set to meet full specified performance.

c. Subject test set to five 48-hour cycles as shown on Standard MIL-STD-170.

d. Make test measurements during the periods specified on the drawing, while the test set is exposed to the humidity conditions shown thereon; in order to determine compliance with full specified performance. Make the measurement not more than 5 minutes after power is applied to the test set and complete them on rapidly as possible.

4.7.6 Vibration test, (See 3.6.1). The test set, with and without its transit case, shall be mounted on the vibration table. During the vibration test, the test set shall be energized and monitored except when installed in its transit case.

4.7.6.1 Variable frequency vibration.

a. Test 1. The test set shall be vibrated with the frequency varying between 10 and 33 cycles per second at a total excursion of 0.006 ±0.006 inch, the frequency varying uniformly from 10 to 33 cycles per second and returning to 10 cycles per second in approximately 1/2 minute.

b. Test 2. The test set shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 0.03 ±0.006 inch, the frequency varying uniformly from 10 to 55 cycles per second and returning to 10 cycles per second in approximately 1 minute.

c. Application of the vibration specified in Test 1 and Test 2 shall be in each of the following directions:

(1) Horizontally, parallel to the major horizontal axis of the test set, for a period of 15 minutes for each frequency range.

(2) Horizontally, at right angles to the major horizontal axis, for a period of 15 minutes for each frequency range.

(3) Vertically, for a period of 15 minutes for each frequency range.

4.7.6.2 Fixed frequency vibration. The test set shall then be vibrated for a period of 3 minutes at each of the four most severe resonant frequencies at the same amplitude as specified in 4.7.6.1 and in the plane observed to give the most severe reaction.

4.7.6.3 Inspection. The test set shall be inspected to ascertain that no fixed parts have become loose or damaged. The test set shall be operated to demonstrate that the performance complies with full specified performance.

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4.8 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physcial interchangeability requirement of 3.10. When a listed dimension is not within specified design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions effect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate together, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

(e) Location of connectors, locking pins, fasteners, slides, and mountings on plug-in assemblies.

(f) Size and form of special threads.

4.9 Procurement inspection of preparation for delivery. The minimum inspection performed by the contractor shall be as specified in 4.9.1 and 4.9.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specified actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.9.1 Preservation and packaging. Inspection of preservation and packaging shall be as specified in Specification MIL-P-116.

4.9.2 Packing and marking of exterior containers. Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be 4 percent and the AQL for minor defects shall be 10 percent.

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4.9.2.1 Inspection lot. A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.9.2.2 Procedure in case of failure. If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.9.2.3 Disposition of nonconforming product. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.10 Rough-handling test (preparation for delivery). When rough-handling test is required by the contract (see 6.2(e)), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling: accuracy of indicated power (see 4.6.5).

4.11 Acceptance inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish	3.3.5
Marking	3.3.6

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Level A. Each Radio Frequency Wattmeter TS-118( )/AP shall be packaged in accordance with Items 1 through 5 of Figure 1 and Table VII.

5.1.2 Level C. Each Radio Frequency Wattmeter TS-118( )/AP shall be packaged as specified in 5.1.1 except Items 3 and 4 shall be Class 2, Item 5 shall be Class 1 and 200 PSI, and the tape in Item 5 shall conform to Specification UU-T-111, Class 2. The tape for the body joint of the box shall be omitted.

5.1.3 Package testing. Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling test and the cyclic exposure test will be required only when invoked in the bid request or contact (See 6.2).

5.2 Packing.

5.2.1 Level A. Each Radio Frequency Wattmeter TS-118( )/AP shall be packed in accordance with items 6, 7 and 8 of Figure 1 and Table VII. The box closure shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. Each Radio Frequency Wattmeter TS-118( )/AP shall be packed as specified in 5.2.1 except item 7 shall be Class 1 and shall be closed with six pieces of tape conforming to Specification UU-T-111, Class 2 - 2 pieces 31-1/2 by 3 inches and 4 pieces 20 by 3 inches.

5.2.3 Level C. Radio Frequency Wattmeter 75-118( )/AP shall be packed in shipping containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. Radio Frequency Wattmeter TS-118( )/AP is a load-resistor-type absorption wattmeter used for measuring average radio frequency (rf) power within the frequency range of 20 mc to 1,400 mc. The wattmeter is used with a selected thermocouple to make power measurements; to aid in the tuning of transmitters; or when used without a thermocouple, as a dummy load.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and amendment.

(b) Type required.

(c) Level of packaging and level of packing required for shipment (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection: Preproduction pack(s) as follows:

a. Makup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon--including rough-handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on figure 1 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required.

(i) Quantity of tools and running spare parts required.

(j) Submission of the statement of treatment as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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6.3 Nomenclature. The parentheses in thenomenclature will be deleted or replaced by a letter identifying the particular design; for example: TS-118W/AP. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information (see 1.1).

6.4 Dimensional data. Sizes of packaging materials prescribed in Section 5 are based on the dimensions of the equipment cited on the applicable Bill of Material. When the dimensions of the equipment vary from those cited, the sizes of the packaging materials shall be adjusted accordingly. When shown in the Bill of Materials (See Section 5), corrugated fiberboard manufactured with A, B or C fluting may be used at the option of the contractor. When the fluting used is not the same as that cited in the Bill of Material, the dimensions of the affected packaging and packing materials will be adjusted accordingly.

Custodians:
Army - EL
Air Force - 11

Reviewer:
Army - EL
Air Force - 85-67-11

User:
Army - MU, MI

Preparing activity:
Army - EL

(Project 6625-0254)

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PACKAGING AND
PACKING DETAILS

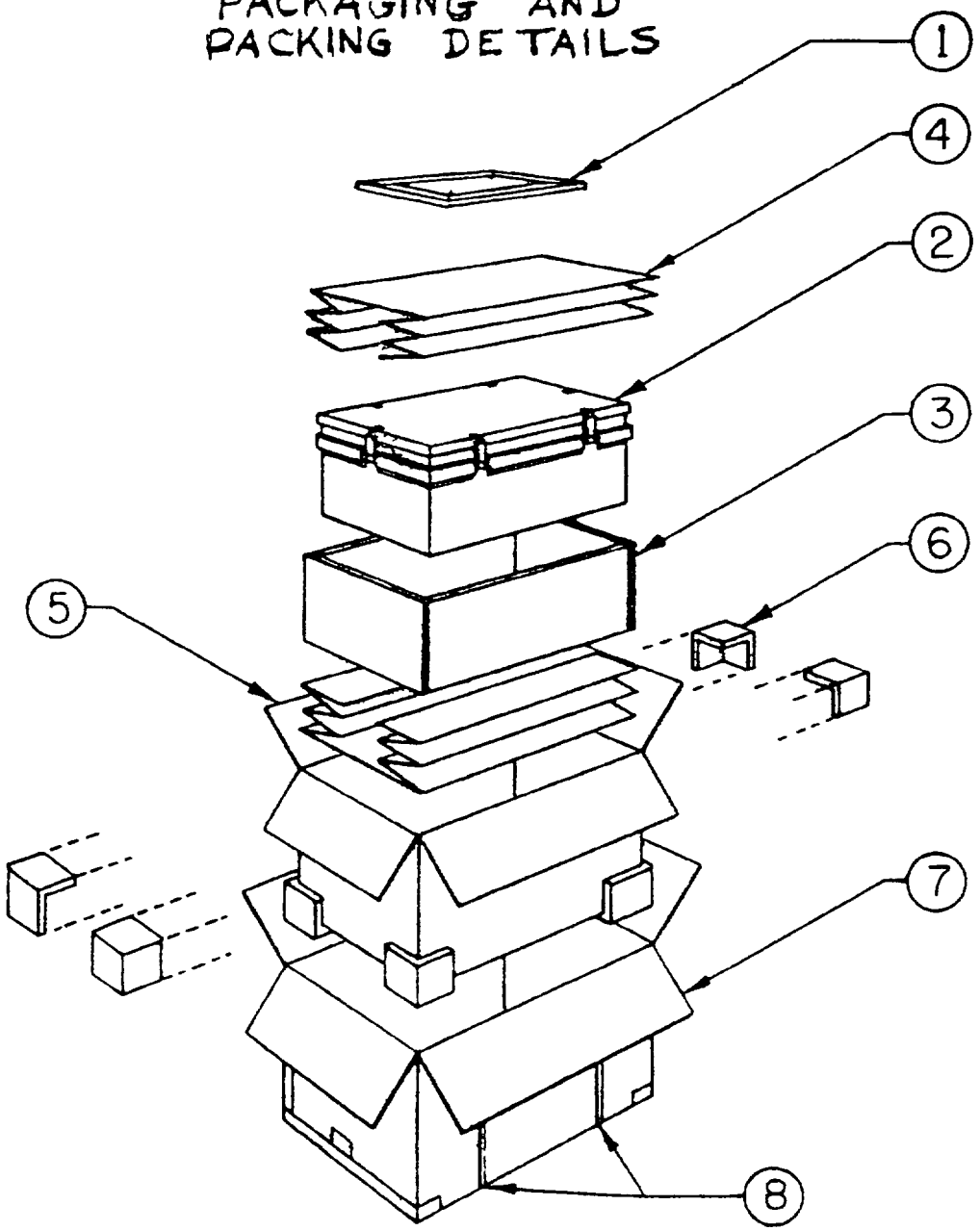


FIGURE 1

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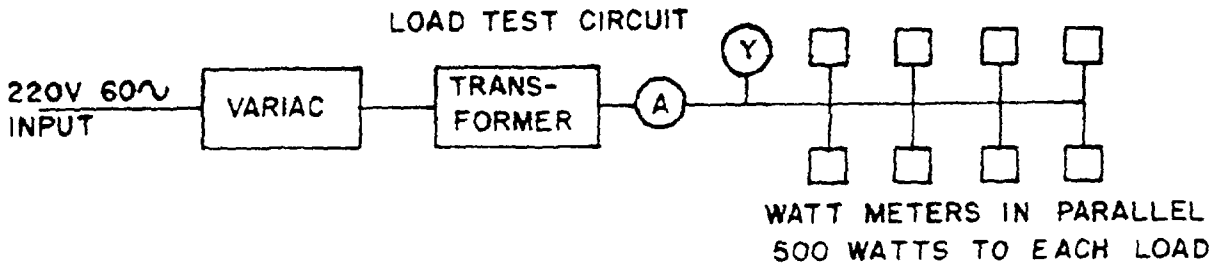


FIG 2

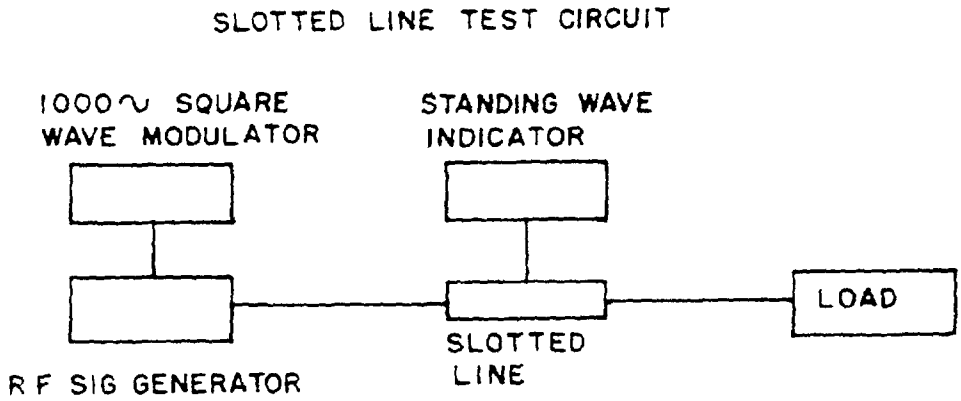


FIG 3

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SENSITIVITY TEST CIRCUIT DIAGRAMS

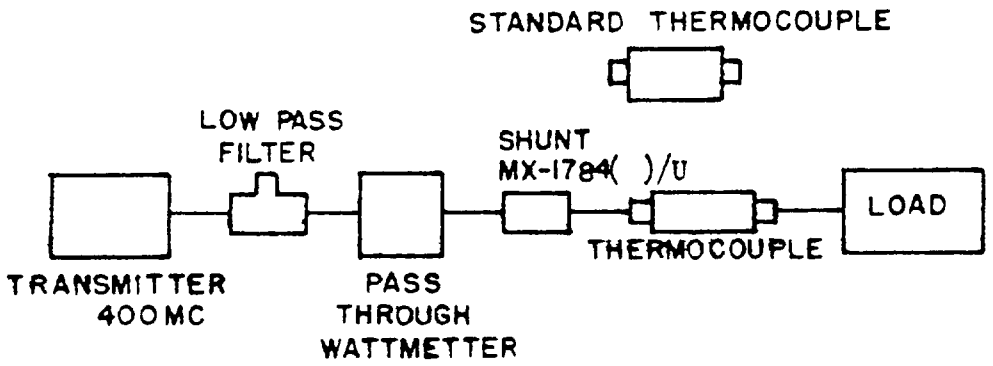


FIG 4A

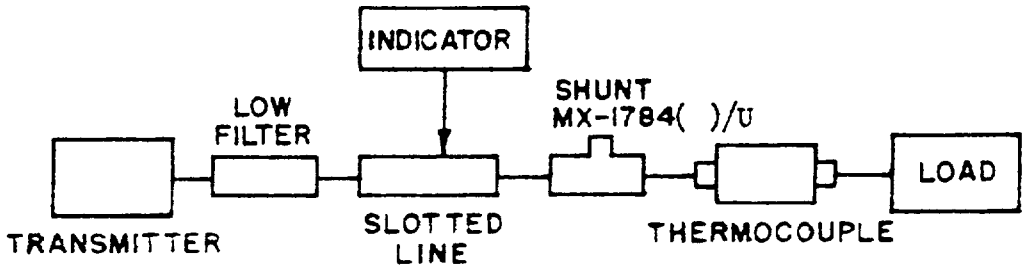


FIG 4B

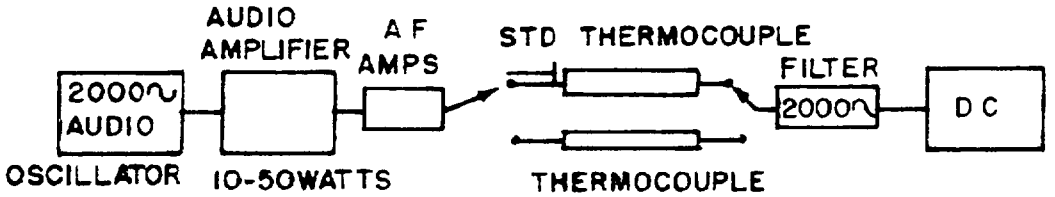


FIG 4C

FIG 4

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VSWR TEST CIRCUIT

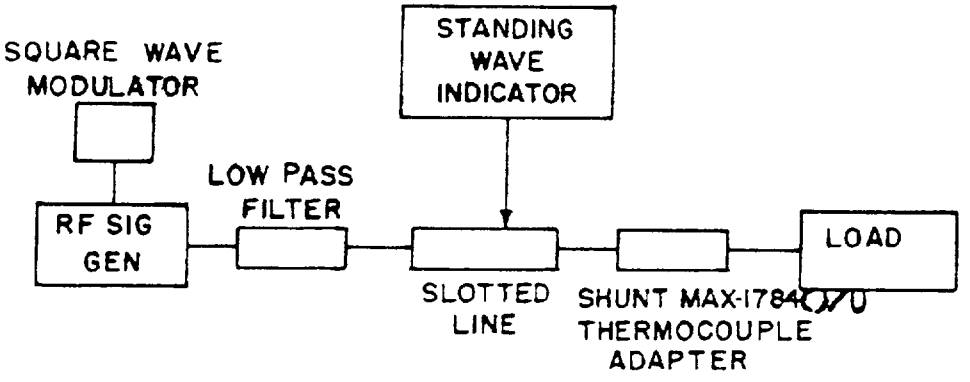


FIG 5

R F POWER TEST CIRCUIT (ACCURACY)

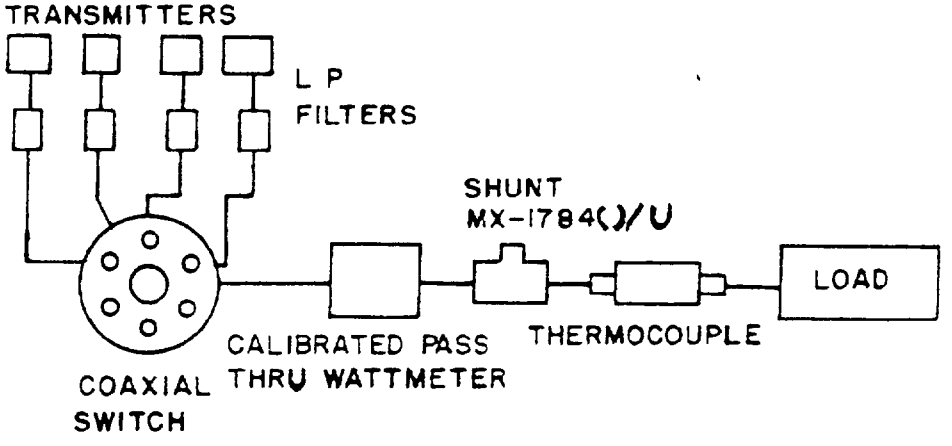


FIG 6

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Table VII. Bill of material

Ref No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec. No.	T	G	C	S	Flute	PSI
1	Technical Manual	2	IC-6										
	Bag, Barrier	2		11-1/2	9		MIL-B-117	11		d			
2	Transit Case CY-1764( )/UPM	1	IC-5	22-1/2	10-7/8	10	Bare Weight with items - 50 lbs.						
	Dummy Load, Electrical DA-140( )/AP	1	)										
	Thermocouples		)										
	MX-1781( )/U	1	)										
	MX-1782( )/U	1	)										
	MX-1783( )/U	1	)										
	MX-1785( )/U	1	)										
	Adapters RF		)										
	UG-83( )/U	1	)	-Place parts in space provided in Transit Case CY-1764( )/UPM									
	UG-213( )/U	1	)										
	UG-999( )/U	1	)										
	Shunt, Tuning		)										
	MX-1784( )/U	1	)										
	Cord CG-92 ( )/U	1	)										
	Twine	3		5	16 ply		T-T-871	1					
3	Sleeve, Butt Joint	1		22-1/2	10-7/8	10	PPP-F-00320	1	1	3		Variety	C
4	Pads, Top and Bottom	2		22-1/2	11	3/4	PPP-F-00320	1	5	3		Variety	A
5	Box, Fiber	1		22-3/4	11-1/4	11-1/2	PPP-B-636	1	3	2		RSC	C
	Tape, Pressure-Sensitive	2		29	3		PPP-T-76						
		4		18	3								
		1		11-1/2	3								

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Table VII. Bill of material - Cont'd

Ref No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES			REQUIREMENT						
				L	W	D	Spec. No.	T	G	C	S	Flute	PSI
6	Corners, Rubber	8		5 Cutout	5 Wall Thickness 0	5 1-1/4 inches	MIL-R-17252	11		2			
7	Box, Fiber	1		25	13-1/2	14-1/4	PPP-B-636	1	3	2	RSC	C	
8	Tape, Filament	2		66	3/4		PPP-T-97						

LEGEND

T - Type
G - Grade
S - Style
C - Class

PACKED WEIGHT 64 lbs.
PACKED VOLUME 2.94 Cu. Ft.

ESC FM 2614 65

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